

Rongxuan Mu 穆荣轩 🖐️

XR+AI HCI Designer / Creative Technologist / Indie Game Developer

Senior HCI Design Specialist @ PICO XR, Bytedance

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Professional Summary

XR+AI HCI Designer and Prototyper with proven experience in **XR OS and games**, grounded in robust **HCI methodologies** that span both hardware and software interaction frameworks. Leverages an **architecture background** to bring a refined **spatial design intuition** to immersive solutions. Passionate about exploring **AI integration** to push the boundaries of user-centric experiences and interactive environments.

Work Experience

Jan 2024 - Now Senior HCI Design Specialist @PICO XR, Bytedance

- Explore innovative methods to improve efficiency and user comfort in general XR OS interaction and specific utilitarian scenarios with AI usage.
- Definition and prototyping of next-generation XR interaction framework, composing and coordinating variable interaction modalities, including eye gaze, hand tracking, speech, XR controller, ring, watch, and traditional pointer mouse with keyboard.
- Support specific HCI solutions for OS built-in Apps and the Developer Platform, such as basic interaction event details like Canvas Drawing, Drag and Drop, Rotate, and Scale.
- Lead and manage the prototyper team for quickly responsive prototypes and iterations.
- Host a series of internal sharing sessions, engaging 30+ industry pioneers.
- Organize routine review of design and development based on cross-functional cooperation.

June 2023 - Jan 2024 HCI Designer and Prototyper @PICO XR, Bytedance

- Optimized Gaze-Pinch interaction with complex parameter-based filtering strategies.
- Designed and optimized direct hand tracking interaction, including direct pinch and poke.
- Redefined mouse and keyboard interactive controls to XR OS with spatial manipulation techniques and features, delivered sound prototypes in cross-team communication.
- Built mathematical diagram-based prototypes for UI parameter calibration; enabled responsive and efficient co-work between the design and engineering teams.
- Conducted a long-term XR interaction performance evaluation of an end-to-end test.

March 2022 - June 2023 Technical Designer @PICO XR, Bytedance

- Designed and developed gameplay in the *2023 Chinese New Year MR showcase*.
- Designed and prototyped an MR multiplayer shooting game, *Matrix Arena*.
- Prototyped sets of interactive properties for a VR live platform.

Aug 2021 - Feb 2022 Real-estate Digital Product Manager Intern @DT-AIOT, Longfor Ltd

- Product design and UX drawing of the internal cross-platform Building Information Model (BIM) app, *Interior Product Selection*, mainly focused on interface renewal and feature iteration.
- Practiced agile product management and coordinated with software engineering teams;
- Participated in the redesign of the internal architectural design management tool.

Side Projects

STRUCTURALISM - 2024

An XR construction game with intuitive hand tracking interaction, vivid physical feedback, and multiplayer cooperation mode. It's also a spatial homage to the 2D game *World of Goo*, with spatial interaction techniques, awarded the Bronze Prize and Best Practice of RTE in the *Rokid Spatial Joy XR Competition*.

OpenBrush - 2023

Contributing to the porting of OpenBrush to the PICO headsets for specific features such as Video Pass-Through and controller adaptation.

Dunhuang VR - 2022

Reconstruction of Dunhuang cave 61 mural of Mount Wutai in a digital virtual space and gamification of the knowledge-based narration, paper accepted by the *International Journal of Human-Computer Interaction*.

Skills

- **Language:** English (Proficient), Mandarin (Native)
- **XR Prototyping:** Unity experiences on Quest, PICO, Apple Vision Pro, Magic Leap, Xreal, Rokid
- **Programming:** Unity C#, Python, Swift, Arduino
- **Design Tools:** Figma, Blender, Rhino+Grasshopper

Education

- **Tianjin University** - Bachelor's degree, Architecture
- **University of California, LA** - TJU+UCLA Joint Studio

Awards

- **2024 Rokid Spatial Joy XR Competition** - Bronze Prize, Best Practice of RTE - **Structuralism**
- **2024 H1 PICO Star Team** - Star Team Contributor - **PICO OS PanoScreen Team**
- **2023 Rokid Developer Competition** - Best Practice of Unity - **Muzuo XR**
- **2023 PICO Dev Jam** - Silver Medalist - **Muzuo XR**
- **2023 Bytedance Staff Tech Jam** - Silver Medalist, Best Business Potential - **MR Interior Designer**

Publications and Patents

- **Mu, R., Nie, Y., Cao, K., You, R., Wei, Y., & Tong, X.** (2023). *Pilgrimage to Pureland: Art, Perception and the Wutai Mural VR Reconstruction*. *International Journal of Human-Computer Interaction*, 40(8), 2002–2018. <https://doi.org/10.1080/10447318.2023.2271665>
- **Han Rui & Mu Rongxuan.** (2023). *From Architecture to Painting and Digital Media Art*. *Time+Architecture* (05), 36-41. <https://doi.org/10.13717/j.cnki.ta.2023.05.03>
- *Memory Reconstruction: Design of a Rural Visitor Center.* (2023). China: Tianjin University Press. ISBN: 978-7-5618-7229-1
- **20+ Patents are not listed here according to NDA restrictions.**